

Liver receptor homolog-1 is essential for pregnancy

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Successful pregnancy requires coordination of an array of signals and factors from multiple tissues. One such element, liver receptor homolog-1 (Lrh-1), is an orphan nuclear receptor that regulates metabolism and hormone synthesis. It is strongly expressed in granulosa cells of ovarian follicles and in the corpus luteum of rodents and humans. Germline ablation of Nr5a2 (also called Lrh-1), the gene coding for Lrh-1, in mice is embryonically lethal at gastrulation. Depletion of Lrh-1 in the ovarian follicle shows that it regulates genes required for both steroid synthesis and ovulation. To study the effects of Lrh-1 on mouse gestation, we genetically disrupted its expression in the corpus luteum, resulting in luteal insufficiency. Hormone replacement permitted embryo implantation but was followed by gestational failure with impaired endometrial decidualization, compromised placental formation, fetal growth retardation and fetal death. Lrh-1 is also expressed in the mouse and human endometrium, and in a primary culture of human endometrial stromal cells, reduction of NR5A2 transcript abundance by RNA interference abrogated decidualization. These findings show that Lrh-1 is necessary for maintenance of the corpus luteum, for promotion of decidualization and for formation of the placenta. It therefore has multiple, indispensable roles in establishing and sustaining pregnancy.